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Improve exploration in deep reinforcement learning for UAV path planning using state and action entropy

Hui Lv¹, Yadong Chen¹ , Shibo Li¹, Baolong Zhu^{1,*} and Min Li²

¹ School of Information and Automation Engineering, Qilu University of Technology (Shandong Academy of Sciences), Jinan, 250353, People's Republic of China

² Beijing Institute of Control Engineering, Beijing 100190, People's Republic of China

E-mail: baolongzhu@hotmail.com

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Abstract

Despite being a widely adopted development framework for unmanned aerial vehicle (UAV), deep reinforcement learning is often considered sample inefficient. Particularly, UAV struggles to fully explore the state and action space in environments with sparse rewards. While some exploration algorithms have been proposed to overcome the challenge of sparse rewards, they are not specifically tailored for UAV platform. Consequently, applying those algorithms to UAV path planning may lead to problems such as unstable training processes and neglect of action space comprehension, possibly causing negative impacts on the path planning results. To address the problem of sparse rewards in UAV path planning, we propose an information-theoretic exploration algorithm, *Entropy Explorer (EE)*, specifically for UAV platform. The proposed EE generates intrinsic rewards based on state entropy and action entropy to compensate for the scarcity of extrinsic rewards. To further improve sampling efficiency, a framework integrating EE and Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithms is proposed. Finally, the TD3-EE algorithm is tested in AirSim and compared against benchmarking algorithms. The simulation outcomes manifest that TD3-EE effectively stimulates the UAV to comprehensively explore both state and action spaces, thereby attaining superior performance compared to the benchmark algorithms in the realm of path planning.

Supplementary material for this article is available [online](#)

Keywords: UAV path planning, deep reinforcement learning, sparse rewards, exploration algorithm, intrinsic rewards.

1. Introduction

Unmanned aerial vehicles (UAVs) have been widely used in various fields, such as transportation [1], agriculture [2], environmental monitoring [3, 4], and military affairs [5], due to their flexible operation and low cost. Regardless of

the application, autonomous path planning is the prerequisite for UAV task completion. Through autonomous path planning, UAV plans an optimal path between the starting and target point under set constraints. According to previous work, path planning in complex environments has always been a challenge [6]. Path planning methods are often classified into non-learning-based and learning-based ones. Non-learning-based methods include the RRT algorithm [7], A-star algorithm [8], ant colony algorithm [9], and genetic algorithm [10]. These algorithms require some or all information about

* Author to whom any correspondence should be addressed.

the environment before planning and demand high accuracy when executing operations. However, uncertainties such as sensor noise and imperfect motion execution mechanisms are inevitable in the real world. In addition, non-learning-based methods require trajectory optimization in the back end, which greatly increases the computational complexity. Therefore, non-learning-based methods are unsuitable for executing highly complex tasks requiring real-time planning in uncertain environments. Learning-based methods, fortunately, can avoid the above drawbacks and have received more attention. The common learning-based methods among learning-based approaches include deep learning (DL) methods, imitation learning methods, and deep reinforcement learning (DRL) methods. DL methods [11–13], exemplified by convolutional neural networks, offer distinct advantages, commencing with its proficiency in feature learning. These models, when exposed to raw pixel data, autonomously acquire salient features, obviating the necessity for laborious manual feature engineering. Nonetheless, DL is not bereft of shortcomings. It inherently necessitates an ample corpus of labeled training data, which can be arduous to amass in the context of UAV path planning. Additionally, these models may exhibit limited adaptability, especially when confronted with novel scenarios. Their decision-making capability is primarily short-sighted, tending to prioritize immediate actions over the optimization of long-term rewards. Imitation Learning methods [14, 15], in contrast, boast several merits. It relies on expert demonstrations to glean optimal behavior swiftly, facilitating the rapid attainment of proficient performance by mimicking the actions of human or expert agents. Nevertheless, imitation learning is not without its drawbacks. Its efficacy hinges heavily on the quality of the expert demonstrations, as any errors present in these demonstrations can propagate to the learning agent. Additionally, it inherently lacks the capacity to explore and experiment in the environment, limiting its adaptability to novel and unencountered situations. The potential for distributional mismatches between expert data and states encountered during training further complicates the generalization aspect of imitation learning. As for DRL methods, this approach combines both DL and reinforcement learning (RL) techniques, emphasizing that the agent (such as a UAV) learns by actively interacting with the environment. DRL methods offer several advantages [16–20], including real-time adaptability to changing environments, end-to-end decision-making, resource management, and scalability. Such benefits enable DRL to make decisions based on current environmental conditions, optimize resource usage, and integrate with other technologies for greater UAV decision-making capabilities. Therefore, DRL is increasingly applied to UAV path planning. For example, He *et al* [21] trained a UAV using the TD3 algorithm and proposed global and local interpretations. Zhang *et al* [22] improved the network structure of the TD3 algorithm with a dual-stream structure, which effectively enhanced its path-planning results.

Despite the seemingly good results of DRL frameworks, the rewards designed in the above works are so dense that UAVs constantly update their policy based on feedback from

reward signals. However, the rewards of some complex real-world environments tend to be extremely sparse. Without guidance from reward signals, it is difficult for UAV to explore the state space and state-action space sufficiently, which may lead to the failure of task completion. Research showed that adding exploration mechanisms to DRL algorithms could effectively solve the exploration problem in complex environments with sparse rewards [23]. A common exploration mechanism design approach is adding intrinsic rewards to stimulate the agent's curiosity. Generally, rewards from the environment are extrinsic rewards, while those awarded by the exploration mechanism are intrinsic rewards. The various approaches to exploration algorithm design can be broadly categorized into four types [24]. The first is count-based exploration, where states with a higher visit count are less novel. However, the count-based method does not reflect the agent's understanding of state-action pair (s, a) . The second is prediction-based exploration, in which states with a large prediction error are awarded a higher intrinsic reward. Yet, the prediction error decreases as the training steps increase, rendering the agent no longer encouraged to visit certain required states. The third is memory-based exploration, which memorizes a history of past states and assigns an intrinsic reward to the current state based on some measure of its similarity to the history. Nevertheless, its complex architecture and high computational complexity greatly limit its application. Finally, information-theoretic exploration calculates the intrinsic reward based on the degree of reduction in environmental dynamics uncertainty. Due to easy implementation and low computational complexity, it has received much attention.

Although the above approaches have alleviated the problem of sparse task rewards, they have only been validated in Atari and MiniGrid games [25]. Furthermore, these exploration algorithms have not been specifically optimized for UAV path planning, and their limitations may be magnified on UAV platform. For instance, the REVD algorithm [26] exhibits sub-optimal sampling efficiency, potentially impacting the overall effectiveness of the path planning process. NGU [27] introduces too many additional modules, thereby increasing the complexity of the algorithm's implementation. As such, there is a compelling need to devise an exploration algorithm for UAV platforms that is both easily implementable and significantly enhances sampling efficiency.

Inspired by information-theoretic exploration, we propose an exploration algorithm, *Entropy Explorer (EE)*, specifically for UAV path planning in the context of sparse rewards. EE generates stable intrinsic rewards based on state and action entropy, effectively mitigating the issue of insufficient extrinsic rewards. Furthermore, to further enhance sampling efficiency, we present a framework that integrates the EE algorithm with the Twin Delayed Deep Deterministic Policy Gradient (TD3) [28] algorithms. Subsequently, the proposed TD3-EE algorithm is tested in AirSim. AirSim is an open-source, cross-platform simulator developed by Microsoft for autonomous systems, including vehicles and drones. It offers a realistic 3D environment with physics simulation, sensor modeling, and support for various vehicle types and sensors. The

simulation results in AirSim indicate that TD3-EE can encourage UAV to fully explore the state and action space, enabling UAV in various states to accurately select the optimal action, which improves the success rate of path planning. The main contributions of this work are as follows:

- (1) An information-theoretic exploration algorithm, EE, is proposed for UAV path planning, which calculates the intrinsic rewards based on state and action entropy to compensate for the scarcity of extrinsic rewards in the environment.
- (2) A framework combining TD3 with the proposed exploration algorithm EE is proposed. It provides stable intrinsic reward signals in sparse reward environments and adjust the focus of the UAV at different stages. At the beginning of training, the UAV is adjusted to focus on the intrinsic rewards, thus encouraging itself to explore the state and action spaces. After accumulating enough task rewards, the framework allows the UAV to shift its attention to extrinsic rewards to better complete the given task.
- (3) The performance of TD3-EE is evaluated in AirSim. The simulation results show that TD3-EE can encourage the UAV to fully explore the state and action spaces, thus outperforming the benchmarking algorithms [26, 27] and achieving a higher success rate in path planning.

2. Preliminaries

2.1. Markov decision processes

Markov decision processes (MDPs) serve as a mathematical framework for modeling sequential decision-making problems in the field of RL. An MDP represents an environment in which an agent interacts by taking actions based on the current state, receiving rewards, and transitioning to new states according to a set of probabilistic dynamics.

Formally, an MDP [29] consists of the following components:

- (1) A set of states, S , which represents the various possible states that the system can be in.
- (2) A set of actions, A , which represents the possible actions that an agent can take in each state.
- (3) A state transition function, ρ , which specifies the probability distribution over next states given the current state and action.
- (4) A reward function, r , which assigns a reward value to each state-action pair and serves as the objective of the decision-making process.
- (5) A discount factor, γ , which determines the relative importance of future rewards as compared to immediate rewards.

2.2. DRL

DRL represents a subset of machine learning that merges RL principles with deep neural networks. Its fundamental aim is to empower agents to acquire optimal decision-making strategies

directly from unprocessed sensory data. DRL has gained significant attention in recent years due to its ability to handle complex tasks and learn from high-dimensional, unprocessed data. By employing deep neural networks as function approximators, DRL algorithms can effectively model intricate policies or value functions, enabling agents to make informed decisions in dynamic environments.

The core idea behind DRL is to utilize RL techniques, such as the MDP framework, in conjunction with deep neural networks to learn complex mappings between sensory input and actions. Deep neural networks offer substantial representational power, allowing DRL agents to capture hierarchical patterns and dependencies in raw sensory data, such as images or raw sensor readings. This direct learning from raw input eliminates the need for manual feature engineering and enables agents to perceive and interpret the environment more comprehensively. DRL algorithms incorporate various key methodologies to facilitate effective learning. Experience replay is a widely employed technique that involves storing agent experiences in a replay buffer and randomly sampling from it during training. This approach breaks temporal correlations, reduces bias, and enhances sample efficiency. Exploration-exploitation strategies are also crucial in DRL, striking a balance between exploring new actions to discover optimal policies and exploiting learned knowledge to exploit promising actions.

2.3. TD3

TD3 [28] represents an advanced algorithm within the realm of RL. It has been meticulously crafted to surmount the intricate challenges frequently encountered in tasks pertaining to continuous control. TD3 employs an actor-critic architecture, a well-established paradigm in the field of RL. The parameter update process for the actor is as follows:

$$\nabla_{\phi} J(\phi) = N^{-1} \sum \nabla_a Q_{\theta_1}(s, a) |_{a=\pi_{\phi}(s)} \nabla_{\phi} \pi_{\phi}(s). \quad (1)$$

The parameter update process for the critic is as follows:

$$\theta_i \leftarrow \operatorname{argmin}_{\theta_i} N^{-1} \sum (y - Q_{\theta_i}(s, a))^2. \quad (2)$$

The actor network selects actions based on the current state, while the critic network estimates the value function to evaluate the chosen actions. The delayed policy updates mechanism in TD3 updates the policy (actor network) less frequently than the value function (critic network), reducing overestimation bias and enhancing policy learning. Twin critics are employed in TD3, utilizing two separate critic networks to estimate the action-value function. This approach reduces variance and enhances training stability. Target networks, periodically updated with the online networks' weights, are used to stabilize the learning process and mitigate rapid changes in value estimates.

TD3 offers several notable contributions and advantages. It allows the learning of robust policies using fewer environment interactions. The use of twin critics and delayed policy updates

enhances training stability, reducing variance in value estimates. TD3 demonstrates generalizability across different continuous control tasks, making it applicable to various domains. Its ability to handle complex, high-dimensional action spaces is particularly advantageous in real-world scenarios.

3. The proposed method

A sparse reward environment denotes a scenario in RL characterized by infrequent or minimal rewards conferred upon an agent by its environment. In these contexts, a significant portion of state-action pairs elicits negligible or zero rewards, while only a small subset of actions yields noteworthy positive or negative rewards. The intricacies of sparse reward environments pose notable challenges to reinforcement learning algorithms, as the scarcity of feedback impedes the agent's acquisition of an effective policy.

In the current landscape of RL algorithms, including TD3 [28], challenges emerge when these algorithms are deployed in environments with sparse rewards. This can be attributed to the limited availability and unpredictable nature of feedback signals. Taking the TD3 algorithm as an example, in such scenarios, TD3's mechanisms for exploration, including the introduction of action noise, might not effectively steer the agent towards rewarding actions. The scarcity of rewards hampers the learning process by rendering the agent's distinction between optimal actions and those resulting in minimal rewards more intricate. TD3's deferred update mechanism can further defer learning, awaiting sufficient data to revise policies. Moreover, the algorithm can become ensnared in local optima due to the scarcity of rewarding experiences. TD3's capacity for generalization is hampered by the inadequacy of diverse data to inform its learning process. The conundrum of maintaining equilibrium between exploration and exploitation is intricate, as TD3's exploration propensity tends to prioritize the discovery of rare, rewarding actions. While TD3 provides a robust framework, the solution to the sparse reward predicament often entails additional techniques such as reward shaping, curriculum learning, or intrinsic motivation, in order to enhance its performance within such environments.

The approach of integrating intrinsic motivation into DRL algorithms to tackle sparse reward environments is garnering increasing attention due to its ability to mitigate the challenges posed by sparse rewards. Intrinsic motivation, by providing internal rewards driven by curiosity or novelty, encourages agents to explore diverse actions and states, accelerating learning and enabling more effective policy discovery. This approach also enhances the adaptability of agents across various tasks, making it valuable for real-world applications. As intrinsic motivation aligns with human learning principles and facilitates autonomous learning, it has gained traction as a promising means to enhance the efficiency, robustness, and versatility of AI agents in complex, uncertain environments. Within the scope of this research, which centers on alleviating the issue of sparse rewards in UAV path planning through the prism of DRL, we propose an information theory-based

exploration algorithm, called *Entropy Explorer (EE)*, that provides intrinsic motivation for most popular RL algorithms. The fundamental principle underlying EE involves the generation of intrinsic rewards through the utilization of both state entropy and action entropy. This approach is meticulously crafted to counterbalance the paucity of extrinsic rewards that commonly characterizes scenarios entailing sparse reward distributions.

3.1. State and action entropy estimation

3.1.1. State entropy estimation. In information theory, entropy measures the uncertainty of a random variable [30]. In other words, a random variable with a higher entropy has greater uncertainty. Likewise, a state in RL with greater entropy is considered more novel and should be assigned a higher intrinsic reward. Therefore, this study uses state entropy as the basis for intrinsic reward assignment to encourage the UAV to access more novel states. In this paper, we employ Rényi entropy [31] to measure the novelty of a state. Rényi entropy is a family of information-theoretic measures quantifying the uncertainty or randomness in a probability distribution.

Definition 1 (Rényi entropy [31]). Consider a random vector, denoted as X , which belongs to the Euclidean space R^m . This random vector is characterized by a probability density function, denoted as $f(x)$, which is defined concerning the Lebesgue measure over R^m . The set χ , defined as $\{x \in R^m : f(x) > 0\}$, represents the support of this probability distribution. The Rényi entropy, a fundamental concept in information theory, is customarily applied to discrete probability distributions and can be described as follows:

$$H_\alpha(f) = \frac{1}{1-\alpha} \log \int_{\chi} f^\alpha(x) dx, \quad (3)$$

where f is the probability distribution, and α is a parameter determining the order of the entropy.

However, Rényi entropy calculation can be quite complex in high-dimensional observation environments. A simple method is to use a k -nearest neighbor estimator to effectively estimate Rényi entropy [32]. Specifically, given a trajectory $\tau = \{s_0, a_0, \dots, a_{T-1}, s_T\}$ in the RL framework, we can approximate the Rényi state entropy as follows:

$$\begin{aligned} \hat{H}_T^{k,\alpha}(d) &= \frac{1}{T} \sum_{t=0}^{T-1} \left[(T-1) V_m C_k \|y_t - y_t^{k-nn}\|^m \right]^{1-\alpha} \\ &\propto \frac{1}{T} \sum_{t=0}^{T-1} \|y_t - y_t^{k-nn}\|^{1-\alpha}, \end{aligned} \quad (4)$$

where $C_k = \left[\frac{\Gamma(k)}{\Gamma(k+1-\alpha)} \right]^{\frac{1}{1-\alpha}}$, $V_m = \frac{\pi^{\frac{m}{2}}}{\Gamma(\frac{m}{2}+1)}$ is the volume of the unit ball in R^m , and $\Gamma(\cdot)$ is the Gamma function.

Based on the observations above, we define the intrinsic reward based on state entropy as follows:

$$r_{se}^i(s_t) := \|y_t - y_t^{k-nn}\|_2^{1-\alpha}, \quad (5)$$

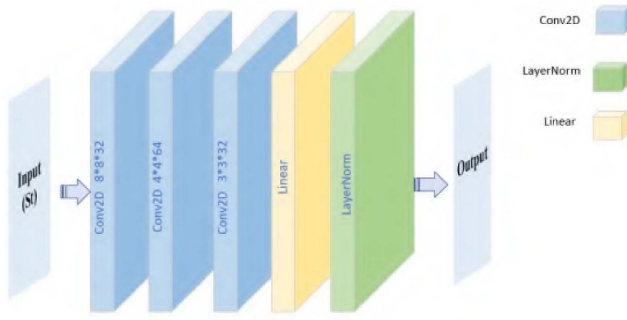


Figure 1. The network structure of the random encoder.

where y_t is the encoding vector of state s_t , y_t^{k-nm} is the k -nearest neighbor of state s_t , $\alpha \in (0, 1) \cup (1, +\infty)$.

Remark 1. The intrinsic reward $r_{se}^i(s_t)$ assigned to a state s_t is higher when the state is more novel or unique. By maximizing the intrinsic reward $r_{se}^i(s_t)$, we encourage the UAV to fully explore the state space, deepen its understanding of the task environment, and avoid planning suboptimal paths due to insufficient environmental information.

To stably and accurately estimate Rényi state entropy, we design a randomly initialized fixed encoder taking advantage of a three-layer convolutional neural network to effectively extract similar features between states. Its specific structure is shown in figure 1.

3.1.2. Action entropy estimation. In the DRL framework, the UAV is controlled by a trained policy network. The UAV observes the current state s_t of the environment, and the policy network outputs a probability distribution. The UAV samples an action from the probability distribution and executes it. If the policy network always outputs a highly deterministic probability distribution, the UAV will not try actions not tried before, thus making it difficult to provide efficient feedback and update the policy. However, the complex task environment and high-dimensional continuous action space in various applications present a considerable challenge for UAV path planning and require a deep understanding of the action space. Therefore, the probability distribution output of the strategy network is not expected to focus on a single action. Instead, the UAV is expected to try different actions to improve the task success rate.

To encourage the UAV to fully explore the action space, we define the intrinsic reward based on action entropy as follows:

$$r_{ae}^i(a|s_t) := - \sum_{a_t} \pi(a|s_t) \log(\pi(a|s_t)), \quad (6)$$

where $\pi(a|s_t)$ represents the probability of taking action a in state s_t .

Remark 2. The intrinsic reward based on action entropy for a UAV increases as the diversity of actions it performs grows. This design can encourage the UAV to fully explore the action

space, thus increasing the diversity of its policy and allowing it to choose the best action to avoid obstacles or plan its path.

3.1.3. Entropy explorer (EE). Based on the above discussion, EE integrates intrinsic rewards based on state entropy estimation and action entropy estimation, which can be expressed as:

$$r^i(s_t) := \log \left(\|y_t - y_t^{k-nm}\|_2^{1-\alpha} - \tau \sum_{a_t} \pi(a|s_t) \log(\pi(a|s_t)) + 1 \right) \quad (7)$$

where y_t is the encoding vector of state s_t , y_t^{k-nm} is the k -nearest neighbor of s_t , $\alpha \in (0, 1) \cup (1, +\infty)$, and $\pi(a|s_t)$ represents the probability of taking action a in state s_t , τ is a non-negative weight factor.

Remark 3. Compared with existing exploration algorithms [26, 27], EE has several potential advantages:

- (1) It has a simple structure that is easy to implement.
- (2) It emphasizes the exploration of both the state and action spaces, providing comprehensive and accurate guidance for the decision-making process.

Therefore, EE is more suitable for UAV platforms to overcome the exploration challenges of sparse rewards.

3.2. Combination with TD3 algorithm

With the objective of augmenting sampling efficiency, this paper introduces a framework that amalgamates exploration algorithms with widely adopted RL algorithms. Within this study, the TD3 algorithm has been chosen as the foundational algorithm to be synergistically combined with the EE algorithm.

Combining the TD3 [28] algorithm with EE offers a potent solution for tackling sparse reward environments. EE can enhance TD3's exploration strategy by providing intrinsic rewards that guide the UAV towards unexplored and informative states. In this approach, the UAV's uncertainty and information gain become driving factors for exploration. By integrating EE with TD3, several benefits arise. First, the UAV becomes more adept at identifying states where its knowledge is lacking, effectively prompting it to explore areas of the state space that might lead to valuable insights. Second, the intrinsic rewards derived from these methods help balance exploration and exploitation, ensuring that the UAV explores while leveraging its learned policy. Moreover, information-theoretic approaches assist in escaping local optima by incentivizing the exploration of unfamiliar territories. This is crucial in sparse reward environments, where the UAV's policy might otherwise become stuck in suboptimal regions. The adaptive nature of these methods allows the UAV to fine-tune its exploration strategy over time, adapting to the changing dynamics of the learning process.

Algorithm. TD3-EE

01: Set up critic networks $Q_{\theta_1}, Q_{\theta_2}$, and actor network π_ϕ with randomly initialized parameters θ_1, θ_2 , and ϕ

02: Establish target networks as follows: θ'_1 is initialized to match θ_1 , θ'_2 is initialized to match θ_2 , and ϕ' is initialized to match ϕ

03: Create the replay buffer $\mathcal{B}$ as part of the initialization process

04: **for** $t = 1$ **to** T **do**

05: Utilize exploration noise to select an action a following the policy $\pi_\phi(s)$, incorporating noise $\epsilon, \epsilon \sim \mathcal{N}(0, \sigma)$

06: Record the received reward r and the resulting new state s'

07: Store the transition tuple (s, a, r, s') in the replay buffer $\mathcal{B}$

08: Randomly sample a mini-batch of N transitions (s, a, r, s') from $\mathcal{B}$

09: $\tilde{a} \leftarrow \pi_{\phi'}(s') + \epsilon, \epsilon \sim \text{clip}(\mathcal{N}(0, \sigma), -c, c)$

10: $y \leftarrow r + \gamma \min_{i=1,2} Q_{\theta'_i}(s', a)$

11: Get a fixed representation $y_t = f_\theta(s_t)$

12: Compute intrinsic reward $r^i(s_t)$

13: Update $\beta_t = \beta_0(1 - \rho)^t$

14: Let $r_t^{\text{total}} = r^e(s_t, a_t) + \beta_t r^i(s_t)$

15: Update critics $\theta_i \leftarrow \text{argmin}_{\theta_i} N^{-1} \sum (y - Q_{\theta_i}(s, a))^2$

16: **if** $t \bmod d$ **then**

17: Update the actor network parameters ϕ using the deterministic policy gradient method:
 $\nabla_\phi J(\phi) = N^{-1} \sum \nabla_a Q_{\theta_1}(s, a)|_{a=\pi_\phi(s)} \nabla_\phi \pi_\phi(s)$

18: Update target networks: $\theta'_i \leftarrow \tau \theta_i + (1 - \tau) \theta'_i$,
 $\phi'_i \leftarrow \tau \phi_i + (1 - \tau) \phi'_i$

19: **end if**

20: **end for**

The structure of the TD3-EE algorithm is shown in figure 2. In TD3-EE, TD3 calculates extrinsic rewards, and EE calculates intrinsic rewards. The rewards of the UAV at time step t are:

$$r_t^{\text{total}} := r^e(s_t, a_t) + \beta_t r^i(s_t), \quad (8)$$

where hyperparameter $\beta_t > 0$ determines the balance between exploration and exploitation. In addition, the proposed framework can adjust the focus of the UAV at different stages. At the beginning of training, the UAV is adjusted to focus on the intrinsic reward, thus encouraging itself to explore the state and action spaces. After accumulating enough task rewards, the framework shifts the attention of TD3-EE the UAV to extrinsic rewards for better task completion. Specifically, we use an exponential decay strategy for β_t , which is calculated as follows:

$$\beta_t = \beta_0(1 - \rho)^t, \quad (9)$$

where ρ is the decay rate, and β_0 is the initial weight.

Remark 4. By combining TD3 with information-theoretic exploration algorithm EE, the UAV acquires the ability to efficiently explore the state space, gain insights from uncertain regions, and escape local optima. This integration provides a robust exploration framework that significantly improves the UAV's ability to learn optimal policies in challenging sparse reward environments.

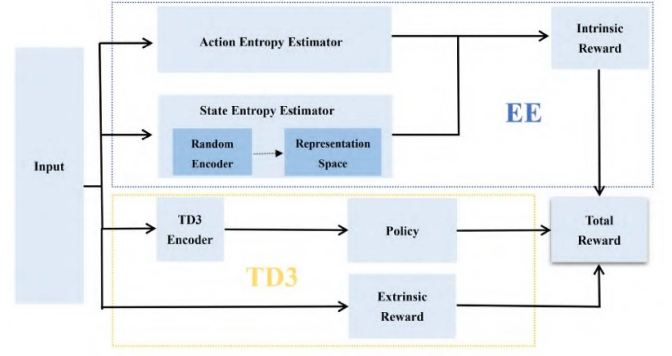


Figure 2. The structure of TD3-EE.

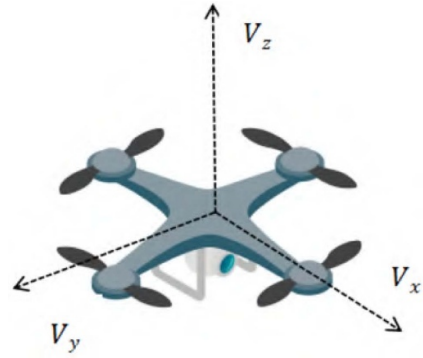


Figure 3. The UAV action space.

4. Experiments and results

Experiments were conducted to answer the following questions:

- (1) Can the proposed TD3-EE algorithm improve UAV performance in sparse reward environments?
- (2) How does TD3-EE compare to benchmarking algorithms?
- (3) Do both the state entropy estimator and the action entropy estimator contribute to the EE algorithm? If so, which component plays a bigger role?

4.1. Problem formulation

4.1.1. Observation space. In DRL, the UAV selects actions based on the observation information from the environment. Thus, designing an appropriate observation space representation is crucial. With the proposed method, the observation space is defined as $o_u = [x, y, z, v_x, v_y, v_z]^T$, where (x, y, z) denotes the position of the UAV, and (v_x, v_y, v_z) denotes its velocity in each dimension, respectively.

4.1.2. Action space. As shown in figure 3, the UAV action space is defined as $a_u = [v_x, v_y, v_z]^T$, where v_x represents the linear velocity of the UAV along the X-axis, i.e. the forward or backward movement of the UAV. In all experiments, we set $v_x = 0.4m/s$ to simplify the training process. v_y represents the linear velocity of the UAV along the Y-axis, i.e. the left or right

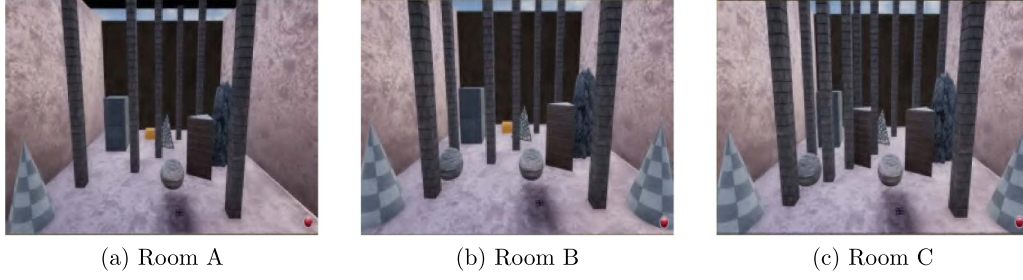


Figure 4. The designed rooms.

Table 1. The number of obstacles in each room.

Shape	Room A	Room B	Room C
Pillar	6	7	10
Cone	1	2	2
Cube	2	2	2
Wedge	1	1	2
Sphere	1	2	2
Tripyramid	1	1	1
Rock	1	1	1
Cylinder	1	1	1
Total	14	17	21

movement of the UAV, and v_z represents the linear velocity of the UAV along the Z-axis, i.e. the upward or downward movement of the UAV. $v_y, v_z \in [-0.6, 0.6] \text{ m s}^{-1}$. The positive direction of the X-axis is in the direction of the first-person view of the UAV.

4.1.3. Reward function. Defining a reward function is important in DRL, as it is used to evaluate the quality of the action taken in the current state and guide the UAV to perform the specified task. To simulate a complex environment with sparse rewards, this study keeps the reward function as simple as possible, which consists of two parts:

- (1) Success reward r^s : the UAV receives a reward of +100 points when exiting the room.
- (2) Collision penalty r^p : the UAV receives a penalty of -10 points when colliding with other objects.

Therefore, the extrinsic reward assigned to the UAV at the current timestep t is:

$$r^e = r^s + r^p. \quad (10)$$

4.2. Simulation environments

To better simulate real-world scenarios, we employed AirSim, a UAV development platform based on Unreal Engine 4. To test the performance of TD3-EE in different scenarios, we designed three rooms labeled A, B, and C. Each room has the same size of $36 \times 36 \times 28$ meters and the same UAV starting position. The only exit in each room is at the top, with the same size. The difference is the number of obstacles in each room

(see figure 4 and table 1), which increases from A to C. Room A has the fewest obstacles, while C has the most. We expect the UAV to find the optimal path to leave the room while avoiding obstacles.

4.3. Comparison with benchmarking algorithms

We compared the proposed EE algorithm with NGU [27], and REVD [26] in the designed rooms. TD3 [28] was used as the underlying RL algorithm for all exploration algorithms, and each experiment was run for 200 000 time steps. Except for each algorithm's hyperparameters, all other settings were kept identical. The algorithms' hyperparameters are listed in table 2.

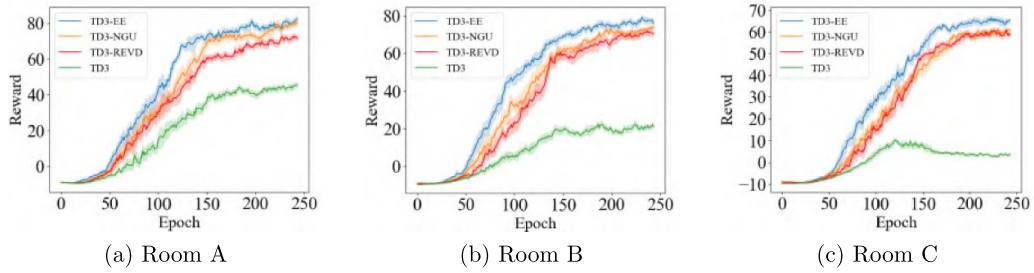
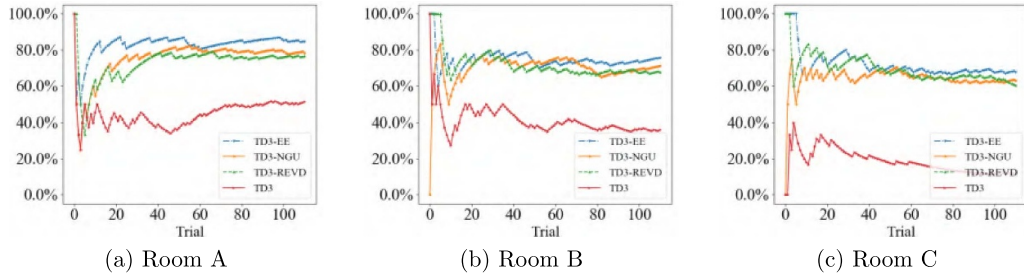
Figure 5 shows the rewards assigned to the UAV in three rooms. Upon training completion, the algorithms were assessed in terms of success rates in rooms A, B and C. A total of 111 trials were conducted per room, and the success rates were computed after each trial. Success rates were determined as the ratio of successful trials to the total. The dynamics of success rates are plotted in figure 6. The success rates after completing 111 trials are presented in table 3. The optimal paths planned by each algorithm are depicted in figure 7. In addition, figure 8 shows the average path length for all successful trials.

Based on the experimental results, the following conclusions can be drawn:

- (1) According to the simulation results in figure 5 and table 3, the UAV trained with the TD3 earned relatively low rewards in all three rooms, indicating that the planned paths were suboptimal or the task success rates were low. As the number of obstacles in the rooms increased, the

Table 2. Hyperparameters of the algorithms.

	Hyperparameter	Value
TD3 [28]:	Learning rate	0.0001
	Policy update delay	2
	Actor exploration noise	0.1
	Target actor noise	0.2
TD3-EE:	K value of k -nearest neighbor (k)	3
	Order (α)	0.1
	Weighting coefficient of $r_{ae}^i(a s_t)$ (τ)	0.0001
	Initial weight (β_0)	0.015
	Decay rate (ρ)	0.001
	Order (α)	0.1
TD3-NGU [27]:	Learning rate	0.0005
	Kernel	0.0001
	Clipping factor	5
	Initial weight (β_0)	0.002
	Decay rate (ρ)	0.002
	Order (α)	0.1
TD3-REVD [26]:	Embedding size (d)	128
	K value of k -nearest neighbor (k)	5
	Order (α)	0.1
	Initial weight (β_0)	0.01
	Decay rate (ρ)	0.001
	Order (α)	0.1

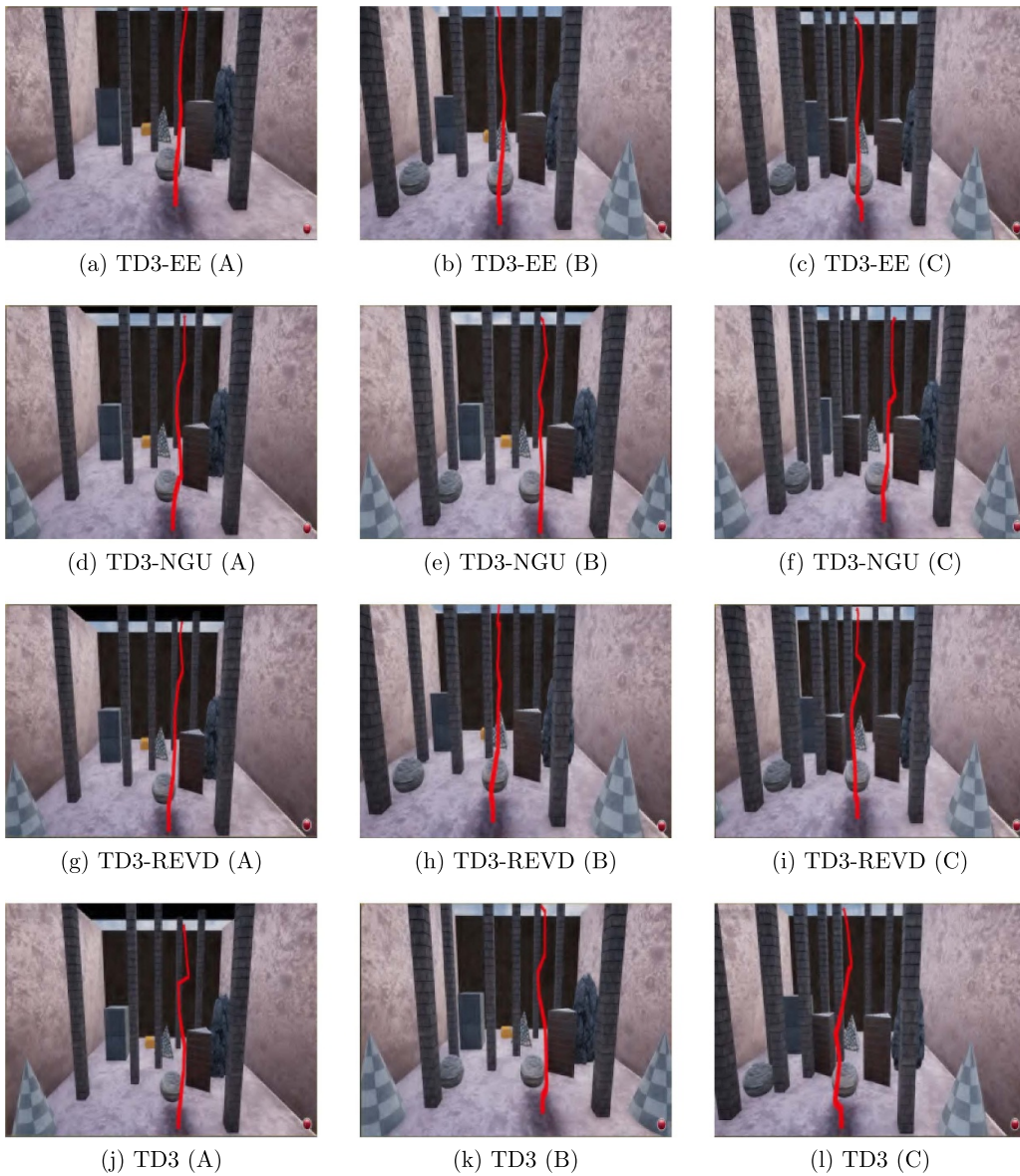
**Figure 5.** The learning curves in each room.**Figure 6.** Success rate variations during the evaluation.

rewards earned by the TD3-trained UAV decreased significantly. Due to the large state and action spaces and the extremely sparse reward function, the policy has difficulty receiving timely feedback. In other words, the TD3 algorithm alone cannot guide the UAV to sufficiently

explore the environment. After combining exploration algorithms such as NGU [27], REVD [26], and the proposed EE, the performance of TD3 improved significantly, indicating that exploration algorithms can effectively address the problem of sparse rewards.

Table 3. Comparison with benchmarking algorithms.

Algorithm	Room	Reward	Path length(m)	Success rate
TD3 [28]	A	45.4 ± 2.8	54.7 ± 3.1	51.35%
	B	21.2 ± 2.2	56.8 ± 2.7	36.03%
	C	4.2 ± 3.1	57.3 ± 3.4	11.71%
TD3-EE	A	82.5 ± 3.0	47.8 ± 2.9	84.68%
	B	75.9 ± 3.4	49.1 ± 2.5	75.68%
	C	65.4 ± 2.2	49.9 ± 3.1	67.57%
TD3-NGU [27]	A	80.3 ± 2.5	50.5 ± 2.3	78.38%
	B	73.8 ± 3.4	53.1 ± 3.8	71.17%
	C	60.5 ± 3.1	52.8 ± 3.7	63.06%
TD3-REVD [26]	A	72.3 ± 2.7	51.2 ± 3.7	76.58%
	B	70.6 ± 3.3	52.7 ± 4.1	67.57%
	C	58.6 ± 2.0	54.6 ± 3.9	60.36%

**Figure 7.** The planned paths in the designed rooms.

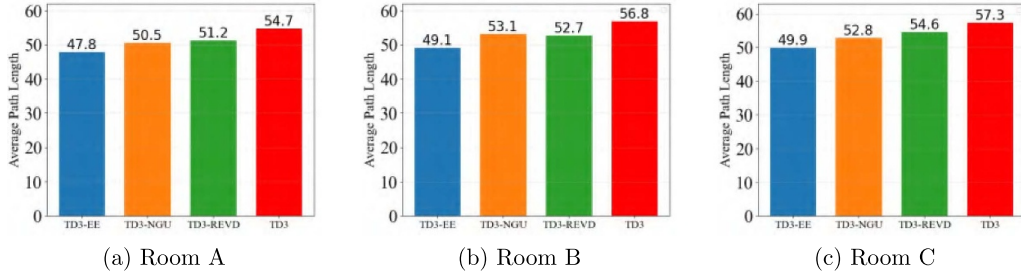


Figure 8. Average path length in each room.

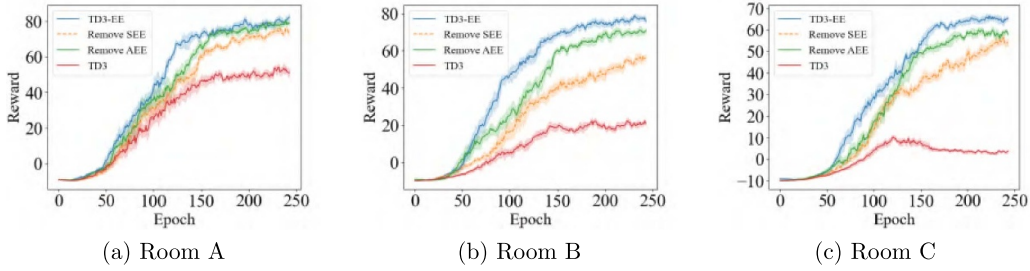


Figure 9. Ablation of intrinsic rewards.

(2) In comparison to the TD3-NGU [27] and TD3-REVD [26], the proposed EE consistently demonstrates superior performance across all three rooms concerning UAV optimization. This superiority is evident across various performance metrics. In terms of rewards, the TD3-EE algorithm attains rewards that surpass the TD3-NGU algorithm by 2.7%, 2.8%, and 8.1% in rooms A, B, and C respectively. In comparison to the TD3-REVD algorithm, the TD3-EE algorithm secures rewards elevated by 14.1%, 7.5%, and 11.6%. With regard to success rates, the UAV's path planning success rates using the TD3-EE algorithm significantly surpass those achieved with the TD3-NGU [27] and TD3-REVD [26] algorithms (as shown in figure 6). Concerning path lengths, the trajectories planned by the TD3-EE algorithm exhibit reductions of 5.3%, 7.5%, and 5.5% in comparison to the TD3-NGU [27] across rooms A, B, and C respectively. Furthermore, when compared to the TD3-REVD [26], the TD3-EE yields shorter paths by 6.6%, 6.8%, and 8.6%. In addition, it is evident that the TD3-EE algorithm exhibits higher sampling efficiency compared to both the TD3-NGU [27] and TD3-REVD [26] algorithms. This remarkable enhancement attributed to the TD3-EE algorithm can be attributed to its unique emphasis on exploration within both the state and action spaces. This comprehensive and accurate exploration affords invaluable guidance in the decision-making process. The synthesis of these exploration aspects leads to significantly amplified outcomes in the context of UAV path planning.

4.4. Ablation of intrinsic rewards

To understand the contribution of both the state entropy estimator component and the action entropy estimator component to the EE algorithm, we conducted a comparative analysis of three variations: the proposed EE algorithm, the EE algorithm with the state entropy estimator (SEE) component removed, and the EE algorithm with the action entropy estimator (AEE) component removed. The experimental results are presented in figures 9–10 and table 4. The dynamics of success rates are depicted in figure 11. In addition, figure 12 shows the average path length for all successful trials.

The findings in this study yield several noteworthy conclusions:

- (1) In the meticulously designed scenarios, the absence of either the SEE component or the AEE component results in a discernible decline in UAV performance. Evidently, the rewards garnered by the UAV in all three rooms exhibit varying degrees of reduction (as illustrated in figure 9). Furthermore, the omission of either module leads to a decrease in the success rate of path planning, as evidenced by figure 11, coupled with an increase in the length of the planned paths, as reflected in figure 12.
- (2) The relative significance of the SEE and AEE components within the TD3-EE framework becomes evident. A comparative analysis reveals that the SEE component exerts a more pronounced influence on TD3-EE. In terms of rewards, the removal of the AEE component incurs reductions of 4.2%, 6.3%, and 11.2% for rooms A, B, and C respectively.

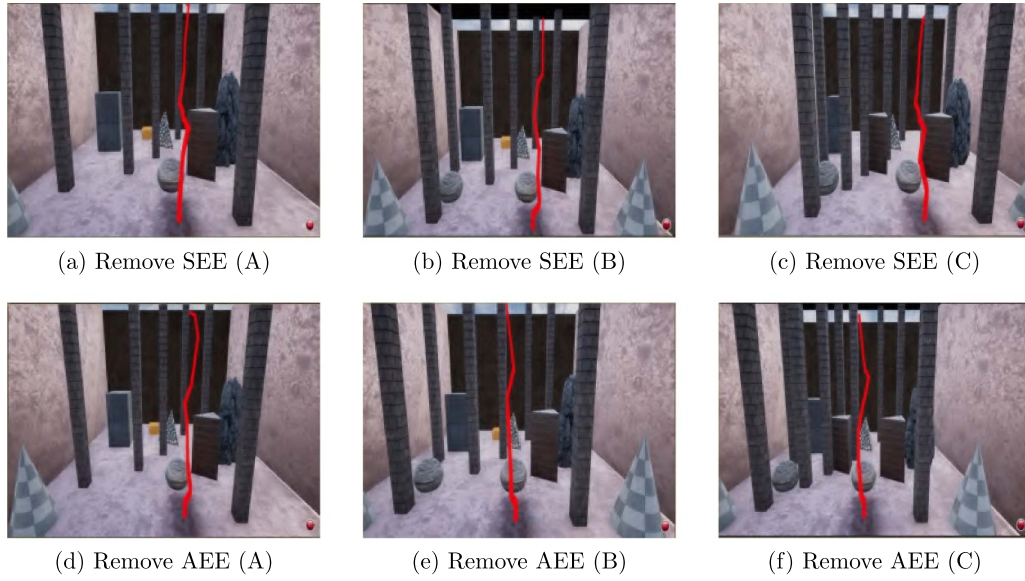


Figure 10. The planned paths in the designed rooms (ablation studies).

Table 4. Ablation of intrinsic rewards.

Algorithm	Room	Reward	Path length(m)	Success rate
TD3 [28]	A	45.4 ± 2.8	54.7 ± 3.1	51.35%
	B	21.2 ± 2.2	56.8 ± 2.7	36.03%
	C	4.2 ± 3.1	57.3 ± 3.4	11.71%
TD3-EE	A	82.5 ± 3.0	47.8 ± 2.9	84.68%
	B	75.9 ± 3.4	49.1 ± 2.5	75.68%
	C	65.4 ± 2.2	49.9 ± 3.1	67.57%
TD3-EE removing SEE	A	73.8 ± 3.2	50.9 ± 2.3	69.37%
	B	55.9 ± 2.1	53.5 ± 2.9	57.66%
	C	53.8 ± 2.4	55.3 ± 3.8	53.15%
TD3-EE removing AEE	A	79.0 ± 3.0	51.5 ± 2.7	74.77%
	B	71.1 ± 1.9	51.2 ± 3.5	63.96%
	C	58.1 ± 2.7	52.8 ± 3.7	55.85%

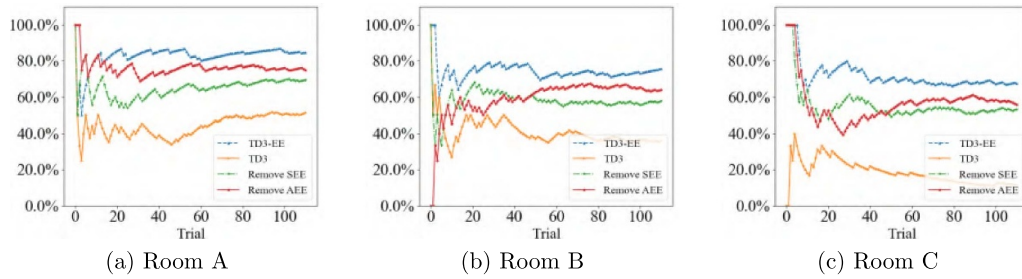


Figure 11. Success rate variations during the evaluation.

C, respectively. In contrast, the elimination of the SEE component corresponds to more substantial reductions of 10.5%, 26.4%, and 17.7% in the respective rooms. Pertaining to the success rate, the AEE component's removal leads to decreases of 9.9%, 11.7%, and 9.7% in rooms A, B, and C, while the SEE component's absence

results in more substantial drops of 15.3%, 18.0%, and 14.4%. Lastly, concerning path length, elimination of the AEE component prompts an increase of 7.7%, 4.3%, and 5.8% in the path lengths for rooms A, B, and C, whereas the removal of the SEE component triggers path length increments of 6.5%, 9.0%, and 10.8%.

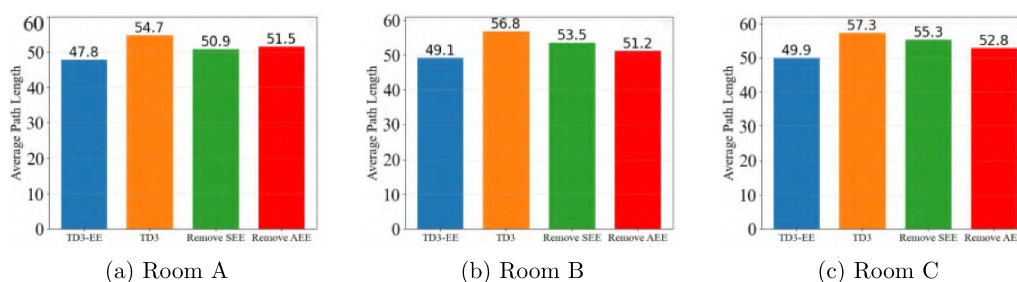


Figure 12. Average path length in each room.

These findings collectively underscore the pronounced importance of discovering new states within the designed sparse reward environment. It is apparent that this aspect outweighs the endeavor of attempting novel actions. We posit that this emphasis on state exploration arises due to the UAV's incentivized exploration of various states, potentially including latent ones, thus facilitating a deeper comprehension of the environment. This comprehension, in turn, emerges as a pivotal determinant influencing the caliber of planned paths.

5. Conclusion and future work

This paper proposed the EE algorithm to solve the sparse reward problem in UAV path planning. EE generated intrinsic rewards based on state entropy and action entropy, enabling the policy to receive meaningful feedback and efficient updates. Furthermore, EE was introduced into the TD3 algorithm to further improve the sampling efficiency. Subsequently, TD3-EE was evaluated in AirSim. The results demonstrated that TD3-EE outperformed the benchmarking algorithms, thus encouraging the UAV to explore the state and action spaces and achieving a higher path planning success rate. Finally, we discussed the contribution of each component on its performance. Overall, our work provides an alternative approach to solving DRL-based UAV path planning in sparse rewards environments.

However, there is still room for improvement in TD3-EE. For example, there are some parameters in our algorithm, which complicates the process of finding the optimal parameters. In the future, we will attempt to design a mechanism for automatic parameter tuning to simplify this process.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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ORCID iD

Yadong Chen  <https://orcid.org/0009-0008-0778-5556>

References

- [1] Chiang W C, Li Y, Shang J and Urban T L 2019 Impact of drone delivery on sustainability and cost: realizing the UAV potential through vehicle routing optimization *Appl. Energy* **242** 1164–75
- [2] Huang H, Tang Y, Tan Z, Zhuang J, Hou C, Chen W and Ren J 2022 Object-Based attention mechanism for color calibration of UAV remote sensing images in precision agriculture *IEEE Trans. Geosci. Remote Sens.* **60** 1–13
- [3] Valente J A, Del Cerro J, Barrientos A and Sanz D 2013 Aerial coverage optimization in precision agriculture management: a musical harmony inspired approach *Comput. Electron. Agric.* **99** 153–9
- [4] Zeybek M 2021 Accuracy assessment of direct georeferencing UAV images with onboard global navigation satellite system and comparison of CORS/RTK surveying methods *Meas. Sci. Technol.* **32** 065402
- [5] Roberge V, Tarbouchi M and Labonté G 2018 Fast genetic algorithm path planner for fixed-wing military UAV using GPU *IEEE Trans. Aerosp. Electron. Syst.* **54** 2105–17
- [6] Pang Z H, Zheng C-B, Sun J, Han Q-L and Liu G-P 2021 Distance- and velocity-based collision avoidance for time-varying formation control of second-order multi-agent systems *IEEE Trans. Circuits Syst. II* **68** 1253–7
- [7] Kala R and Warwick K 2011 Planning of multiple autonomous vehicles using rrt 2011 *IEEE 10th Int. Conf. on Cybernetic Intelligent Systems (CIS)* pp 20–25
- [8] Duchoň F, Babinec A, Kajan M, Beňo P, Florek M, Fico T and Jurišica L 2014 Path planning with modified a star algorithm for a mobile robot 6th *Conf. on Modelling of Mechanical and Mechatronic Systems (MMaMS)* vol 96 pp 59–69
- [9] Zaza T and Richards A 2014 Ant colony optimization for routing and tasking problems for teams of UAVs 2014 *UKACC Int. Conf. on Control* pp 652–5
- [10] Lou T, Chen N, Jiao Y, Zhao H and Zhao L 2023 A consider unscented particle filter with genetic algorithm for UAV multi-source integrated navigation *Meas. Sci. Technol.* **34** 095105
- [11] Thoma A, Stierner L, Braun C, Fisher A and Gardi A G 2023 Potential of hybrid neural network local path planner for small UAV in urban environments *AIAA SCITECH 2023 Forum*

- [12] Padhy R P, Verma S, Ahmad S, Choudhury S K and Sa P K 2018 Deep neural network for autonomous UAV navigation in indoor corridor environments *Proc. Comput. Sci.* **133** 643–50
- [13] Akreimi M S, Neji N and Tabia H 2023 Visual navigation of UAVs in indoor corridor environments using deep learning *Integrated Communications, Navigation and Surveillance Conf. (ICNS)* pp 1–6
- [14] Harris C M, Choi Y and Mavris D N 2021 Imitation learning for UAS navigation in cluttered environments *AIAA Scitech 2021 Forum*
- [15] Gao F, Wang L, Zhou B, Zhou X, Pan J and Shen S 2020 Teach-Repeat-Replan: a complete and robust system for aggressive flight in complex environments *IEEE Trans. Robot.* **36** 1526–45
- [16] Tong G, Jiang N, Biye L I, Xi Z, Ya W and Du W 2021 UAV navigation in high dynamic environments: a deep reinforcement learning approach *Chin. J. Aeronaut.* **34** 479–89
- [17] Hu Z, Zhang Y and Huang H 2023 Reinforcement learning for energy efficiency improvement in UAV-BS access networks: A knowledge transfer scheme *Eng. Appl. Artif. Intell.* **120** 105930
- [18] Xu Y H, Sun Q M, Zhou W and Yu G 2022 Resource allocation for UAV-aided energy harvesting-powered D2D communications: a reinforcement learning-based scheme *Ad Hoc Netw.* **136** 102973
- [19] Artizzu C O, Allibert G and Demonceaux C 2022 OMNI-DRL: Learning to fly in forests with omnidirectional images *IFAC-PapersOnLine* **55** 120–5
- [20] Kim M, Kim J, Jung M and Oh H 2022 Towards monocular vision-based autonomous flight through deep reinforcement learning *Expert Syst. Appl.* **198** 116742
- [21] He L, Aouf N and Song B 2021 Explainable deep reinforcement learning for UAV autonomous path planning *Aerosp. Sci. Technol.* **118** 107052
- [22] Zhang S, Li Y and Dong Q 2022 Autonomous navigation of UAV in multi-obstacle environments based on a deep reinforcement learning approach *Appl. Soft Comput.* **115** 108194
- [23] Pathak D, Agrawal P and Efros A 2017 Curiosity-driven exploration by self-supervised prediction *34th Int. Conf. on Machine Learning (ICML)* vol 70 pp 2778–87
- [24] Hao J et al 2023 Exploration in deep reinforcement learning: from single-agent to multiagent domain *IEEE Trans. Neural Netw. Learn. Syst.* **64** 1–21
- [25] Badia A P et al 2020 Agent57: outperforming the atari human benchmark (arXiv:2003.13350)
- [26] Yuan M, Li B, Jin X and Zeng W 2022 Rewarding episodic visitation discrepancy for exploration in reinforcement Learning *36th Conf. on Neural Information Processing Systems (NeurIPS)*
- [27] Badia A P et al 2020 Never give up: Learning directed exploration strategies *8th Int. Conf. on Learning Representations (ICLR)*
- [28] Fujimoto S et al 2018 Addressing function approximation error in actor-critic methods *Int. Conf. on Machine Learning*
- [29] Ruan A, Shi A, Qin L, Xu S and Zhao Y 2020 A reinforcement learning-based Markov-decision process (MDP) implementation for SRAM FPGAs *IEEE Trans. Circuits Syst. II* **67** 2124–8
- [30] Makkuva A V and Wu Y 2018 Equivalence of additive-combinatorial linear inequalities for shannon entropy and differential entropy *IEEE Trans. Inform Theory* **64** 3579–89
- [31] Zhang C and Cai Y 2021 Exploration by maximizing Rényi entropy for reward-free RL framework *Proc. AAAI Conf. on Artificial Intelligence* pp 10859–67
- [32] Leonenko N, Pronzato L and Savani V 2008 A class of Rényi information estimators for multidimensional densities *Ann. Stat.* **36** 2153–82